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THE VIEW AT END OF Q3 2026

The Enterprise GenAI Stack

A new baseline for the enterprise GenAI stack, set by this research: the reference architecture, what to select, what to deliberately not build yet, and how the advice changes for seven kinds of reader.

Reference architecture • 9 layers • 8 enterprise controls • 140 products • 1,449 sources

PEOPLE | INSIGHTS | DECISIONS | A BRIGHTER TOMORROW

The answer in one sentence

Build a firm-owned control and evidence plane first. Run regulated work as deterministic workflows with one bounded model step and a human approval gate. Consume models as a two-vendor portfolio. Treat every product beneath the plane as replaceable.

Control first

Gateway, identity, privacy, configuration, evaluation and one evidence store are the firm's own. They make exit a configuration change.

Workflows, not free agents

Every serious framework now separates deterministic workflows from agent loops. Regulated work defaults to the workflow.

Products are components

Ownership of 'neutral' tools changed hands repeatedly in 2025–26. Independence and exit must be designed in, not assumed.

A new baseline, set by this research

17

layers and controls: nine stack layers (L1–L9) and eight enterprise controls (C1–C8)

140

products profiled: 92 in the stack layers, 48 in the controls; 138 scored

192

high-risk claims re-checked by two adversarial verifiers

1,449

sources logged with URL and access date

How the work was done

- Eight parallel research streams, primary sources first; two adversarial verifiers re-checked the high-risk claims (no fabrications found). The result is the baseline every later edition is measured against.
- Seventeen chapters on a 13-part template, each product scored 1–5 on eight criteria with generic and regulated-FS weights.
- Reader checkpoints decided the scoring rules and the contested tiers; every claim is labelled verified, reported, judgement or recommendation.

Ten findings that shape the stack

1

Catalogue → control system

The eight enterprise controls are where GenAI risk is managed.

3

Models are tiered families

Gated top tiers and short lives: model choice is a lifecycle discipline.

5

Protocols under foundations

MCP and A2A now sit in a Linux Foundation project; agent identity is a product category.

7

Memory layer is thinning

Open editions narrowed; hyperscalers ship memory in their runtimes.

9

Regulatory anchors moved

SR 26-2 excludes GenAI; EU high-risk duties now 2 Dec 2027.

2

Neutral tools were bought

Arize, Langfuse, Promptfoo, Portkey, Lakera, Voyage, Jina and others changed hands.

4

Workflows vs agents

Every serious framework now ships both; this is the key design choice.

6

Vector DB is a feature

Hybrid search ships in most stores; retrieval is a governed index.

8

Supply chain is attack surface

A gateway package was compromised; a tool broker leaked tokens.

10

'Open source' often is not

Source-available, AGPL, non-commercial and threshold licences.

Independence can no longer be assumed from a product's origins



Product (layer)	New owner	Status (as of 8 October 2026)
Arize Phoenix and AX (evaluation)	Dynatrace	Completed 1 October 2026
Langfuse (evaluation)	ClickHouse	Announced 16 January 2026
Promptfoo (red-teaming)	OpenAI	Announced 9 March 2026; closing not published
Voyage AI (embeddings)	MongoDB	Closed February 2025
Jina AI (embeddings)	Elastic	Completed 9 October 2025
Tavily (search API)	Nebius	Closed 19 February 2026
OpenRouter (model router)	Stripe	Agreed 19 August 2026; pending
Portkey (gateway), Protect AI (security)	Palo Alto Networks	Completed 2025–26
Lakera (guardrails)	Check Point	Completed 22 October 2025
Guardrails AI; Helicone	Harvey; Mintlify	Completed; Helicone in maintenance mode

What it means

A model vendor now owns a red-team tool; security vendors own gateways. Effective challenge and exit planning have to be designed, not inherited.

What we do about it

Keep evaluation datasets, routes and policies in Git; keep two red-team tools, one independent of the model vendor; re-run due diligence on every change of control.



The regulatory anchors moved in 2026



17 Apr 2026

2 Aug 2026

18 Mar 2027

2 Dec 2027



SR 26-2 replaces SR 11-7

EU AI Act GPAI enforcement

PRA PS7/26 • FCA PS26/2

EU AI Act Annex III

Generative and agentic AI are out of scope: firms must write their own GenAI standard

Commission powers over general-purpose model providers start

Material third-party notifications and registers begin

High-risk duties deferred by Regulation (EU) 2026/1744

Concentration sits with the clouds, not the model vendors

DORA's first critical-provider list and the UK CTP designations name hyperscalers and no model vendor. PRA SS1/23 and the EU AI Act are now the operative model-risk anchors.





The architecture

Eight hypotheses tested; nine stack layers under one firm-owned control plane of eight controls



Four planes under one firm-owned control plane

CONTROL PLANE (firm-owned)

C1 AI traffic gateway · C2 guardrails · C3 privacy service · C4 identity · C5 configuration of record · C6 FinOps · C7 AI security · C8 governance and evidence store

MODEL PLANE (L1, L2)

Two-vendor portfolio + small + open-weight · in-region model access · vLLM exit route

AGENT PLANE (L3, L4)

Workflows by default · bounded agent steps · durable execution · tools behind a governance sub-layer

KNOWLEDGE PLANE (L5, L6, L7, L8)

Memory service (built last) · derived, entitlement-filtered stores · retrieval optimisation · ingestion envelope

EVALUATION AND OBSERVABILITY PLANE (L9)

Firm OpenTelemetry collector · platform of record · CI and online evaluations · two red-team tools

How to read it

The control plane governs every call and holds the evidence. Planes beneath it are built from replaceable products.

L1–L9 and C1–C8 are the baseline's identifiers in every document, the explorer and the dataset.

Hypothesis verdicts: how evidence shaped the layers



#	Hypothesis	Verdict
H1	Inference is several jobs; the gateway is a control	L2 covers model access, serving and optimisation; the gateway is C1 (model, tool and agent traffic)
H2	Agent framework is not one layer	L3 holds deterministic workflows (default), bounded agent steps and durable execution
H3	Identity and tool governance	Keep: a tool-governance sub-layer in L4; C4 and C7 make the decisions
H4	Memory vs retrieval	Memory is a governed service on the stores layer; build it last
H5	Vector databases are retrieval stores	L6 is 'Retrieval, knowledge and memory stores'
H6	Embeddings + reranking	Together they are L7 'Retrieval optimisation'
H7	Ingestion needs lineage, DLP, ACLs	Keep, widened: a built ingestion envelope around replaceable parsers
H8	Evaluation is cross-cutting	L9 is an evidence plane, sharing one evidence store with C8



Twelve decisions that define the architecture

- 1 One gateway of record for all model, MCP and agent traffic; two deployments, in region, failing closed
- 2 Evaluation and observability from day one; datasets in Git; two red-team tools
- 3 A firm-owned, immutable evidence store keyed by trace ID
- 4 A two-vendor model portfolio in region, plus a small model and a self-hosted exit route
- 5 Deterministic workflows by default, with approval gates only a named human can resume
- 6 Read-only tools for authoritative data, behind a governed tool gateway
- 7 Every agent a registered identity acting on behalf of a human, deny by default
- 8 One privacy service at six enforcement points
- 9 Retrieval inside platforms already run, as a rebuildable, entitlement-filtered index
- 10 Git as the configuration of record for prompts, pins and policies
- 11 Capability separation as the first security control
- 12 Memory last, as a governed record class



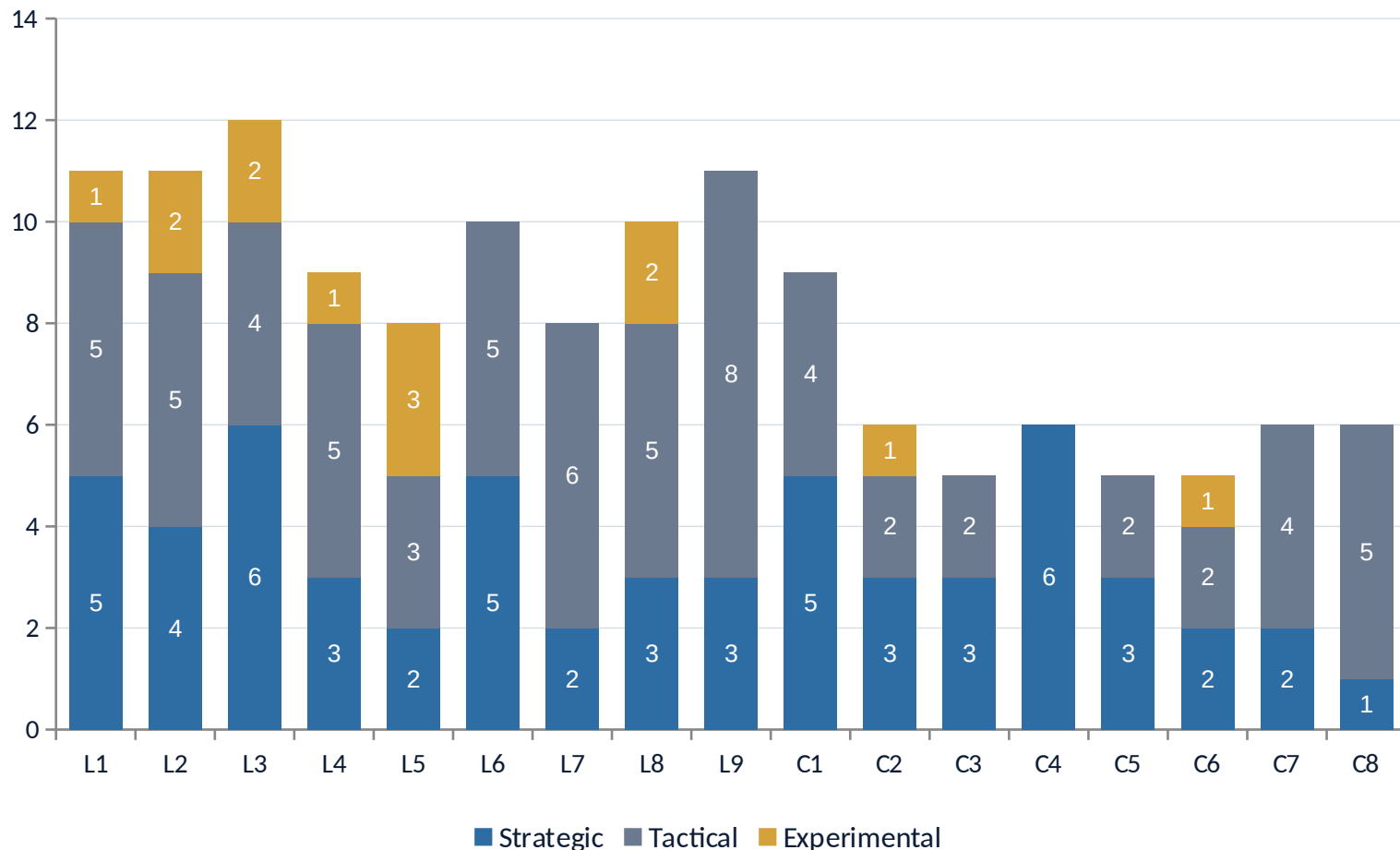
What to select

138 products scored on eight criteria · 58 Strategic · 67 Tactical · 13 Experimental



Where the Strategic choices are

Products by tier, layer (L1→L9) and control (C1→C8)



A Strategic tier carries a condition

18 of the 58 apply only where a given cloud is primary, or where a platform is already operated. The condition is the decision.

What drives the tiers

Regulated-FS weights favour deployment flexibility and low lock-in, so open, self-hostable components take most Strategic slots. No commercial governance tool reached Strategic on public evidence.

The cloud-neutral core we would select today



Layer / control	Selection	Independent alternative or note
L1 models	Two vendors from OpenAI, Anthropic, Mistral, Gemini; Gemma 4 or Mistral self-hosted	For Claude: GPT-6.1 Sol, Gemini 3.8 Flash or Mistral Medium 3.5
L2 inference	Primary cloud's in-region model service + vLLM exit route	SGLang as backup engine once its CVE fix is confirmed
L3 orchestration	LangGraph on Temporal	Microsoft Agent Framework, ADK, Pydantic AI
L4 tools	Read-only MCP tools behind a governed gateway	OpenAPI tools (MCP originated at Anthropic)
L6 / L7 retrieval	pgvector (or Elasticsearch where run) + Sentence Transformers	Qdrant or Milvus after a failed load test
L8 ingestion	Docling + Unstructured inside a built envelope	Specialist parsers for hard documents
L9 evaluation	Langfuse or MLflow + firm OpenTelemetry collector	Two red-team tools, one vendor-independent
C1 gateway	LiteLLM Enterprise (hardened, pinned) or Kong AI Gateway	Cloud gateway where that cloud is primary
C3 privacy / C4 identity	Presidio behind a privacy API • workforce IdP + OPA + SPIFFE	Cloud DLP / Entra Agent ID per cloud
C5 configuration	Git as configuration of record; prompts as code	Vendor registries are caches only



Treat models as a portfolio, not a bet

Primary mid-tier

One vendor through the primary cloud's in-region model service

Second mid-tier

An unrelated vendor qualified on the same evaluation suite

Small model

Classification, routing and judging at low cost

Open-weight exit route

Gemma 4 or Mistral on vLLM: the stressed-exit drill

- Top tiers are gated and lifetimes are short: one Gemini Flash version lives about five months. Re-qualification is routine work, not an event.
- Chinese-origin families: decide the route, not the brand. Self-hosted weights by explicit policy or in-tenant hosting; never the vendor's own API for client data.
- Numbers are never generated by the model; they come from the attribution engine through read-only tools.

The new baseline: 17 layers and controls, 140 products



Nine stack layers

- L1 models as a pinned portfolio
- L2 access, serving, optimisation
- L3 workflows and bounded agents
- L4 tools behind a governed gateway
- L5 memory as a governed record class
- L6 retrieval stores
- L7 retrieval optimisation
- L8 ingestion envelope
- L9 evaluation and observability

Eight enterprise controls

- C1 AI traffic gateway
- C2 guardrails as policy
- C3 privacy service
- C4 agent identity and authorisation
- C5 configuration of record
- C6 AI FinOps
- C7 AI security
- C8 model risk, governance and evidence store

Facts as at Q3 2026

- Current lines: Gemma 4, Z.ai GLM-5.x, Mistral Medium 3.5
- archived or closing: TGI, OpenAI Agent Builder; Helicone in maintenance
- not verifiable: EthicalAgents, Ragoos
- a router is not the access layer; desktop runtimes stay out of production



Deliberately not selected, on evidence

Avoid	Why
Archived or deprecated: TGI, OpenAI Agent Builder, AutoGen for new code, Helicone, Zep Community Edition	Archived, shut down, superseded or in maintenance mode
EthicalAgents, Ragoos	Could not be verified as real products
A third-party broker holding client tokens (Composio managed cloud)	Unresolved token-exposure incident, May 2026
Jina weights, MinerU above thresholds, a modified AGPL Firecrawl server	Licence blockers without a commercial licence
LM Studio as a service; training-on-data model tiers	Terms blockers

Moved to monitor by the reader (CP5): Chinese-origin vendors' own APIs for client data (the route rule still applies) and billing intermediation. Not avoided but not yet: memory, autonomous agents with write tools, multi-agent delegation, fine-tuning, own-GPU fleets, dedicated vector databases without a failed load test.



The regulated firm's view

What supervisors will ask for, and the worked example that makes it concrete



What a supervisor or validator will ask to see

Model risk (SS1/23)

A GenAI model-risk standard, inventory entry, independent validation report, monitoring results, change records

EU AI Act

Use-case classification, AI literacy records, Article 26 logs (six months or more) and oversight assignments for any high-risk use

Third parties (PS7/26, DORA)

Provider assessments, notifications from 18 March 2027, register entries, tested exit plans including a stressed exit

Data

Processing-region evidence per call, transfer risk assessments, erasure runbooks covering vectors, caches and backups

Audit trail

Per-output evidence pack: prompt and model versions, data snapshot, tool calls, evaluation results, approver identity

Security (OWASP 2026)

Red-team results mapped to the LLM and Agentic Top 10 lists; supply-chain gate records

Worked example: the attribution-commentary agent

1 Analyst request

Identity and policy check (C4)

2 Workflow graph

Deterministic; one drafting step (L3)

3 Read-only data

Attribution engine via governed tools (L4)

4 House style

Retrieval of prior commentary (L6–L8)

5 Draft

Model via the gateway, in region (C1, L2, L1)

6 Evaluate

Every figure matches the engine (L9, C2)

7 Approve

Named portfolio manager (C4)

8 Evidence pack

Stored, immutable, reproducible (C8)

A multi-asset fund, Brinson-style attribution (allocation, selection, currency). Generic and illustrative: not any firm's platform.

What the agent must never do matters more than what it can do



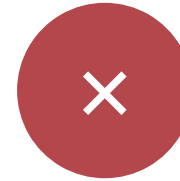
Generate or alter a number

Every figure comes from the approved attribution output and is traceable



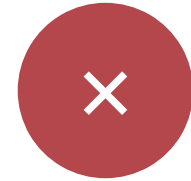
Present inference as data

Commentary distinguishes the engine's figures from the narrative



Publish without approval

Only a named, entitled person can resume the approval gate



Leave no record

Outputs are reproducible; evaluation evidence is retained

The control that catches the one failure that matters

A wrong number in a client commentary is caught by a deterministic numeric-faithfulness check: an evaluation scorer reused inline as a guardrail.





Delivery

Four reference stacks · build vs buy · lock-in · an 18-month roadmap



Four reference stacks: Stack A leads

A • Regulated enterprise

LEAD

Private deployment, governance and evidence before speed; tested exit for every material dependency

B • Cloud-native managed

The primary cloud's lead service in each category; firm-owned evidence and configuration keep exit possible

C • Open-source first

Open licence and self-hosted route for every component; the cloud as infrastructure, not platform

D • Minimal start-small

Stack A cut to one use case, with an explicit 'do not build yet' list

Same control-plane minimum in every stack

Gateway, evaluation harness, evidence store, privacy API, agent identity, Git configuration, budgets that fail closed. Stacks differ in who runs the components, not in whether they exist.

Build the control statement; buy the commodity



Component	Decision	What the firm owns
Models (L1)	Buy; adopt open weights	Portfolio policy, qualification suite, retirement calendar
Stores (L6)	Reuse what is run	Retrieval interface and rebuild pipeline
Ingestion (L8)	Build the envelope	No product emits lineage; parsers adopted
Evaluation (L9)	Hybrid	Datasets and scorers in Git; platform replaceable
Gateway (C1), guardrails (C2), privacy (C3)	Hybrid	Routes, policies, test sets, keys; the product is replaceable
Identity (C4)	Buy the IdP, build the policy	Agent registration, policies in Git
Configuration (C5), FinOps (C6)	Build (thin)	Release manifest, evaluation gate, allocation rules
Governance (C8)	Build the store, buy the workflow	Evidence store and inventory outlive any tool
Fine-tuning	Do not build yet	May turn a deployer into a provider under the AI Act

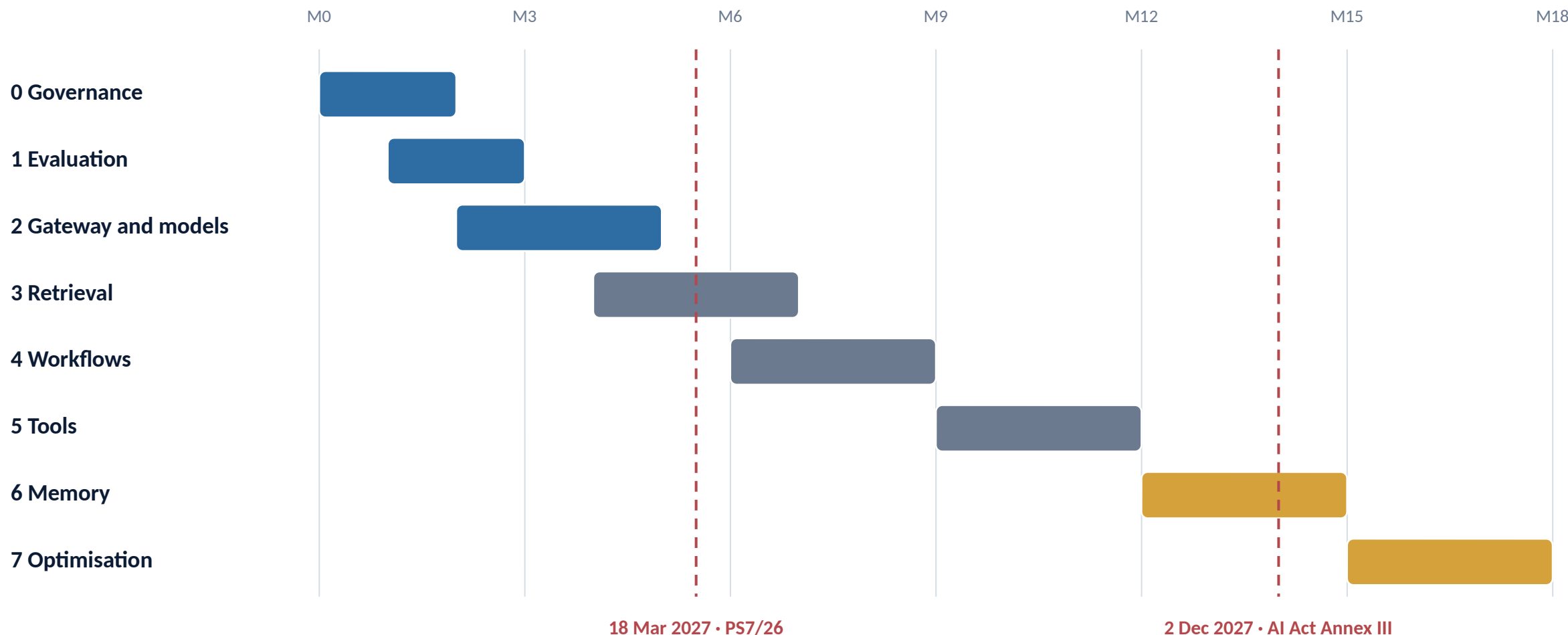


Which lock-in is acceptable, and which is not

Area	Acceptable	Unacceptable without an exit
Models (L1)	Open weights the firm holds	A single proprietary vendor for an important service
Orchestration (L3)	Open frameworks	Vendor-hosted agent state for regulated workflows
Tools (L4)	MCP and A2A as open specifications	Credential custody in a third party's multi-tenant cloud
Stores (L6)	A vector index rebuildable from source	Entitlement logic only in proprietary syntax
Evaluation (L9)	Production APM module	Datasets only in a vendor UI
Configuration (C5)	Pins in a manifest	Prompt text or change history only in a vendor registry
Governance (C8)	Policy packs, OpenLineage	An evidence store held only by a vendor

The pattern: own the data, the policy and the evidence; let the engines be bought and replaced.

An 18-month roadmap, governance first and memory last



October 2026 to March 2028. Phases overlap; each starts when its predecessor's exit criteria are in sight. The exit drill (model switched by configuration only) must pass before March 2027.



Why evaluation comes before the first model call

1

It is the validation evidence

Without a suite there is nothing for a validator to re-perform and nothing to approve

2

It is what makes exit possible

A replacement model is qualified only by running the same suite

3

Every later change is a model change

Model, prompt, embedding and tool changes all need a regression baseline

4

Evidence cannot be recreated

Free tiers keep traces for weeks; an unrecorded output has no reproducible record



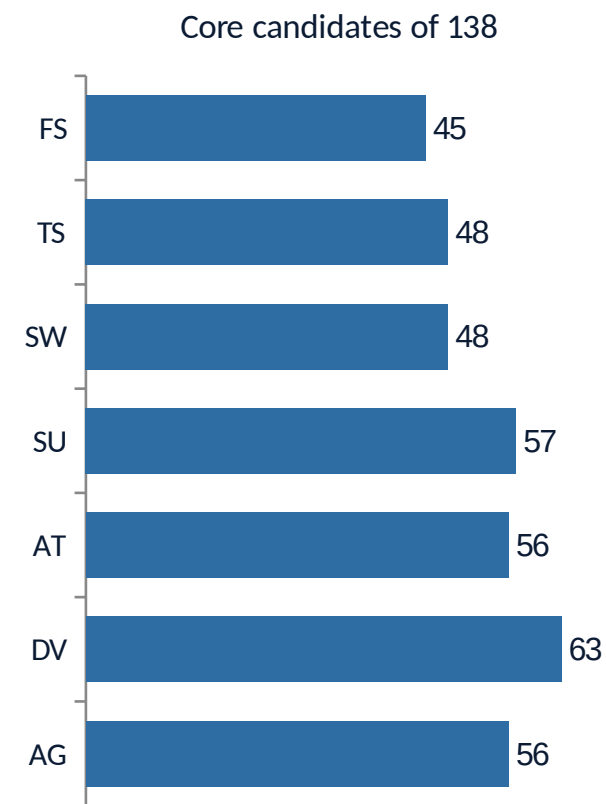
One stack, seven lenses

The same 138 scored products and criterion scores, re-weighted for six further readers (Parts XIII–XVIII)



Seven readers, one stack, different weights

View	Reader	Weighted most	Part
FS	Regulated asset manager deploying GenAI for itself	Security and compliance 20%	Parts I–XII
TS	Runs GenAI inside services it operates for customers	Reliability 15% and cost 15%; lock-in 5%	Part XIII
SW	Ships GenAI inside software customers install and run	Deployment flexibility 20%	Part XIV
SU	Early-stage company building an AI-native product	Technical 25%, cost 20%	Part XV
AT	Start-up selling AI tools into the enterprise stack	Technical 20%; ecosystem 15%	Part XVI
DV	Start-up selling agentic SDLC tools to developers	Technical 25%; ecosystem and cost 15% each	Part XVII
AG	Start-up selling agents to enterprises	Technical 20%; enterprise, security, deployment, ecosystem 15% each	Part XVIII



Same facts and criterion scores; only the weights change

A core candidate scores at least 3.6 under the view's weights, has no criterion at 1 and is not Experimental. The master tiers and their conditions still stand.

Service provider: tenant-aware, metered and sold



TS

The same control plane as the regulated firm, but every element is tenant-aware, metered and sold. Tenant isolation, cost per call and service levels replace model-risk validation as the hardest problems.

Write a tenancy standard first

Tenant, user and agent in every token; filters injected server-side; every cache, index and store addressed by tenant; a cross-tenant leak test blocks every release.

Design the margin in

Every model call is cost of goods sold: batch, prompt caching and a small model in every feature; cost per call reported per tenant and feature against price.

Make the second vendor a live capacity route

Sized and drilled for real load, because discount tiers throttle and one tenant's misuse can suspend the whole service.

Sell a service, not a pass-through model API

Flow vendor terms down to tenants, list every model route as a subprocessor, and ship an AI assurance pack and NIS2 and CRA incident runbooks.

Read more: Part XIII of the master document · core candidates under this view: 48 of 138 (FS 45)



Software company: own the release plane



SW

The vendor ships into many control planes it does not own. It is a manufacturer and provider: its product plugs into each customer's gateway, identity, telemetry and evidence tools.

Publish a support matrix

At least two unrelated hosted vendors, reached through the customer's own account or gateway, plus one Apache 2.0 or MIT model bundled for air-gapped sites.

Route through the customer's gateway

One configurable model endpoint; no provider SDKs in feature code; no vendor-run endpoint on the request path of a self-managed product.

Gate every shipped component

Redistribution, not use, decides what ships: a licence register and an AI-aware SBOM per release; signed, safetensors-only bundled weights.

Treat the CRA and PLD as deadlines

Vulnerability reporting since 11 September 2026; software a PLD product from 9 December 2026; CRA main obligations from 11 December 2027.

Read more: Part XIV of the master document · core candidates under this view: 48 of 138 (FS 45)



Start-up: speed first, exits kept open cheaply

SU

A fortnight, not a quarter, on the control plane. What is built in the first two weeks decides how cheaply the company can change its mind later.

Build four things in the first fortnight

One pinned gateway with a key and budget per customer; a tenant ID on every row, vector and trace; standard tracing; an evaluation set in Git with the prompts.

Let credits pick the first vendor, not the last

Credits mostly pay only for the credit-giver's models: qualify a second vendor on the same evaluation set, switchable by configuration.

Isolation before model risk

Keeping one customer's data out of another's answers is the first control problem; the users are customers' staff, so use customer identity, not a workforce directory.

Two things do not bend

Pin the gateway: a widely used open-source gateway shipped poisoned releases in March 2026. Keep evaluation sets and prompts as the company's own IP.

Read more: Part XV of the master document • core candidates under this view: 57 of 138 (FS 45)

AI-tools start-up: easy to adopt, easy to leave



AT

The buyer's architecture already assigns the product's place: a replaceable call-out inside the buyer's control plane, running where the data is and writing evidence into the buyer's stores.

Publish the integration contract

Gateway hooks with a latency budget and typed outcomes; spans to the customer's OpenTelemetry collector; verdicts to its evidence store; usage in an open cost format.

Run where the buyer's data is

One build as EU SaaS, private-endpoint SaaS and a container in the customer's account; no hidden model dependency; policy as code, not console settings.

Beat the free baseline on evidence

Open tools and incumbents' bundles already ship the feature: win on recall, operations or evidence, never on having the feature.

Pass diligence and plan for a change of owner

Many 'neutral' tools were acquired: subprocessor list, DORA addendum, exit plan and change-of-control notice; the start-up's own CRA duties apply too.

Read more: Part XVI of the master document · core candidates under this view: 56 of 138 (FS 45)



Agentic SDLC: trust is earned in the pull request

DV

The crown jewels are source code and the credentials that change it. Two questions decide the deal: where does our code go, and can we see everything the agent did?

The customer's model endpoint by default

Every call through the customer's gateway; two unrelated model vendors qualified per task; retention reported route by route, because it depends on the model.

The pull request is the boundary

Own branch, isolated sandbox with default-deny egress, short-lived repository-scoped tokens; the agent never merges, a named human does.

Fill the audit gap

An evidence record of every model call, tool call and command, written to the customer's store, with pinned OpenTelemetry spans.

Sell depth, neutrality and evidence

Incumbents ship breadth and the enterprise baseline. Win on one task done well; cache stable prefixes, the main cost lever.

Read more: Part XVII of the master document • core candidates under this view: 63 of 138 (FS 45)

Agent start-up: an ungoverned agent is not bought

AG

The buyer governs every agent through its own control and evidence plane. The start-up sells domain logic, a tested workflow and an evaluation suite, not a second identity plane, gateway or evidence store.

Run under the customer's identity

Registered in the customer's directory with a sponsor; on-behalf-of tokens per tool audience; no standing secrets; revocation within 15 minutes.

Ship the strict profile of incomplete standards

Tool authorisation on, signed agent cards, short-lived audience-bound tokens, no sub-delegation; tools read-only by default, through the customer's gateway.

Keep nothing the customer must audit

Model calls only to a customer-supplied endpoint; one evidence record per output in the customer's store; workflow, prompts and autonomy budget as files.

Treat governance fit as the product

Incumbents are building agent platforms. The test: an evidence pack for every output, revoke by one identity, switch model by one route.

Read more: Part XVIII of the master document · core candidates under this view: 56 of 138 (FS 45)

What holds across all seven lenses

A named person approves

Anything consequential: a support reply, a merged pull request, a payment run. The model drafts; a person decides.

Memory comes last

Long-term memory of customer or client content waits; when built, it is a governed record class with erasure by person.

Configuration in source control

Prompts, model pins and policies released through review, in Git, as one manifest.

An evidence record for every output

Keyed by a trace ID, in a store the firm or its customer owns, never only in a vendor's.

A pinned, signed supply chain

A five-person team is as exposed to a poisoned package as a bank.

Every model call through a gateway

The firm's own or the customer's: one route to switch a model, enforce a budget and record the call.

What changes is who owns the control plane, who pays for inference and which rulebook applies: deployer, operator, manufacturer, provider or supplier. Every lens keeps a second, unrelated model vendor qualified on the same evaluation set; only the reason for it changes.

Decisions for the leadership team



Adopt the reference architecture

Control plane first; Stack A as the target; the lock-in rules as design policy

Name the accountable owner

An SMF for GenAI model risk and a GenAI standard to SS1/23 quality

Fund Phases 0-2 now

Governance, evaluation harness, gateway and two-vendor model access, with the exit drill before 18 March 2027

Choose the first use case

The attribution commentary: clear boundaries, checkable numbers, a named approver





Voyan®

THE VIEW AT END OF Q3 2026

Own the control plane. Rent the components.

Full evidence: master document (seven views, Parts I–XVIII), product technical appendix (138 scored products, fit by view), offline explorer with a view selector, dataset and source archive. Every claim is labelled and sourced. Researched and drafted with an Anthropic model; Anthropic items scored on the same rubric, with tiers set by the reader marked.

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DECIDE
BUILD
LEARN
ADAPT

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TO EVOLUTION

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